

NFL DRAFT VALUE MODEL

Which Combine Metrics Actually Matter?

Combine dataset: 2000–2026 · 8,968 prospects across 10 position groups

Scope: position-specific athletic benchmarks, a draft-value prediction model, a custom Scouting Athletic Index (SAI), historical player comparables, outlier/freak-athlete detection, drill-importance-by-position analysis, draft steals/reaches, and position clustering into athletic archetypes.

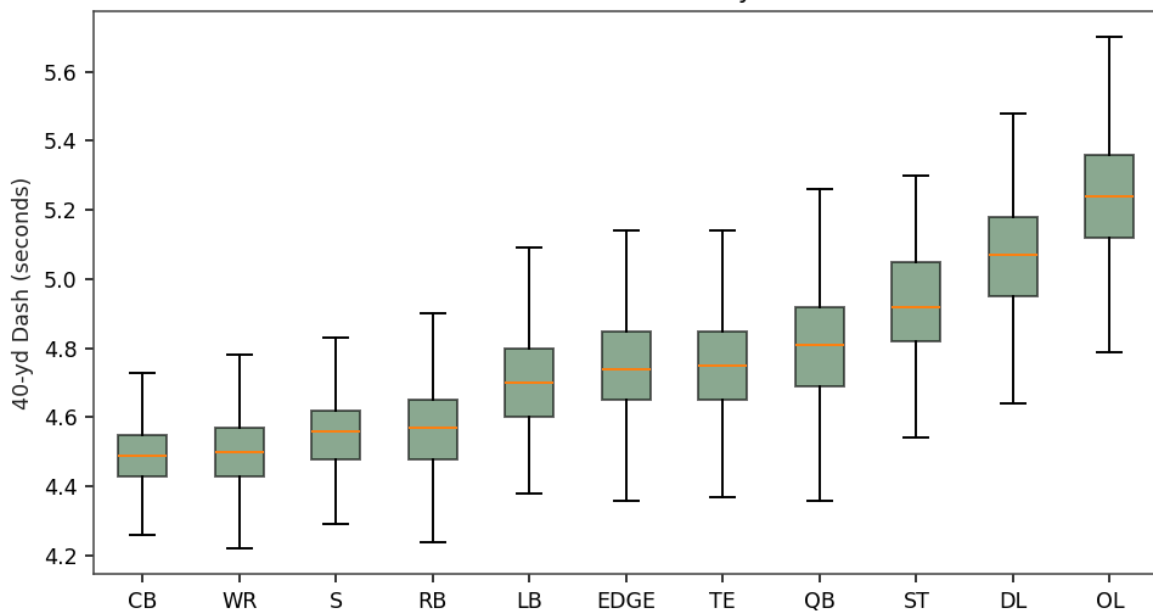
Headline finding: combine measurables alone explain only about 21% of the variance in where a player is drafted ($R^2 \approx 0.21$). Workout numbers matter, but tape, production, and intangibles matter more — which is itself a finding, not a limitation to hide.

Part 1 — Position-Specific Athletic Profiles

Instead of one blended combine average, every position gets its own elite / average / poor benchmark, built from position-specific percentiles (90th/10th, direction-aware so lower-is-better drills like the 40 are handled correctly).

Position	Elite 40yd	Avg 40yd	Elite Vert	Elite Broad	Elite Bench
QB	4.59	4.813	35.5"	120.7"	24.1
RB	4.41	4.58	38.5"	125.0"	26.0
WR	4.37	4.505	39.5"	129.0"	20.0
TE	4.581	4.757	37.5"	123.0"	26.0
OL	5.01	5.245	32.5"	112.0"	32.0
EDGE	4.56	4.749	38.5"	125.0"	29.7
CB	4.37	4.492	40.0"	130.0"	19.8
S	4.42	4.554	39.5"	128.9"	22.0

40-Yard Dash Distribution by Position

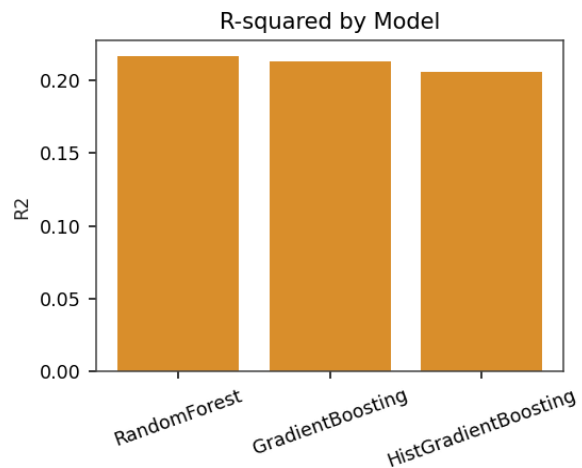
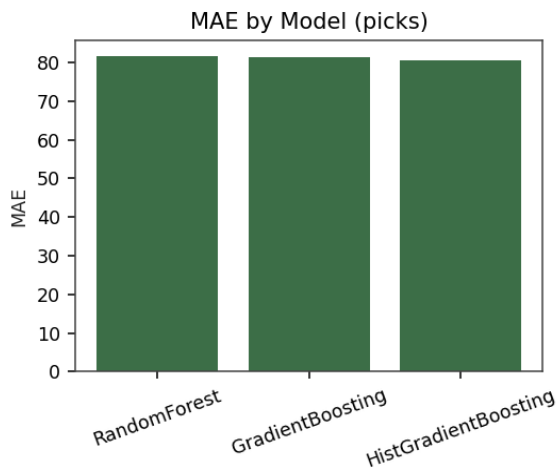


40-yard dash distribution by position — skill positions (WR/CB/S/RB) cluster fast, line positions (OL/DL) run much slower, exactly as scouting intuition predicts.

Part 2 — Draft Prediction Model

Random Forest, Gradient Boosting, and HistGradientBoosting (scikit-learn's gradient-boosted-tree family, used in place of XGBoost/LightGBM since this sandbox has no network access to install them — drop-in swap locally with the same fit/predict API) were trained to predict draft value (overall pick number, with undrafted players slotted beyond the last real pick of their class).

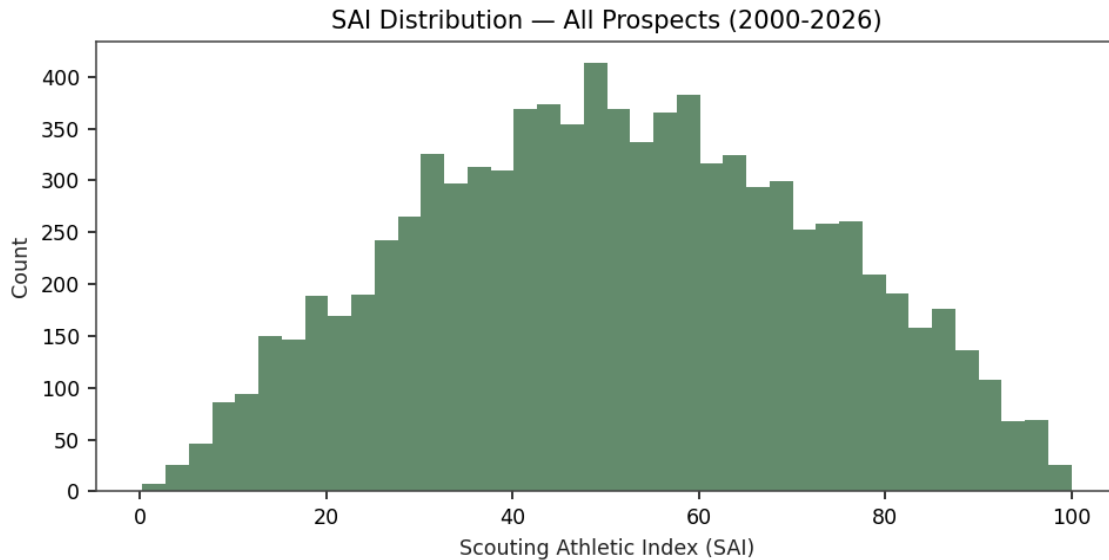
Model	MAE (picks)	RMSE	R ²
RandomForest	81.51	94.8	0.2163
GradientBoosting	81.21	95.0	0.213
HistGradientBoosting	80.58	95.43	0.2059



Best model by MAE: **HistGradientBoosting**, off by about 81 picks on average. Global permutation importance: **40-yard dash and weight dominate**, with position, agility drills, and bench press contributing much less.

Part 3 — Scouting Athletic Index (SAI)

SAI is an original 0–100 composite (not a clone of RAS): four physics-grounded sub-scores — **Speed Score** (weight-adjusted 40 time), **Explosiveness** (weight-adjusted vertical+broad), **Agility** (inverse cone+shuttle), and **Power/Size** (weight-adjusted bench, falling back to a BMI-style size term when bench is missing, since bench has ~40% missingness in this dataset) — each converted to a within-position percentile, then blended with position-specific weights (a corner's SAI weights speed/agility more; an offensive lineman's weights power/size more).



SAI is centered near 50 by construction (percentile-based), with a long tail of true freak athletes above 90.

Part 4 — Historical Comparables

Given a player, the model finds the closest historical athletes at the same position using three interchangeable distance metrics: Euclidean, cosine similarity, and Mahalanobis distance (which accounts for correlation between drills). Example outputs from the engine:

Travis Hunter (CB, 2025) — Euclidean

Mansoor Delane (2026) · Derek Stingley (2022) · Benjamin Morrison (2025) · Ashton Youboty (2006)

Calvin Johnson (WR, 2007) — Euclidean

Michael Jenkins (2004) · Vincent Jackson (2005) · Michael Floyd (2012) · Savion Williams (2025)

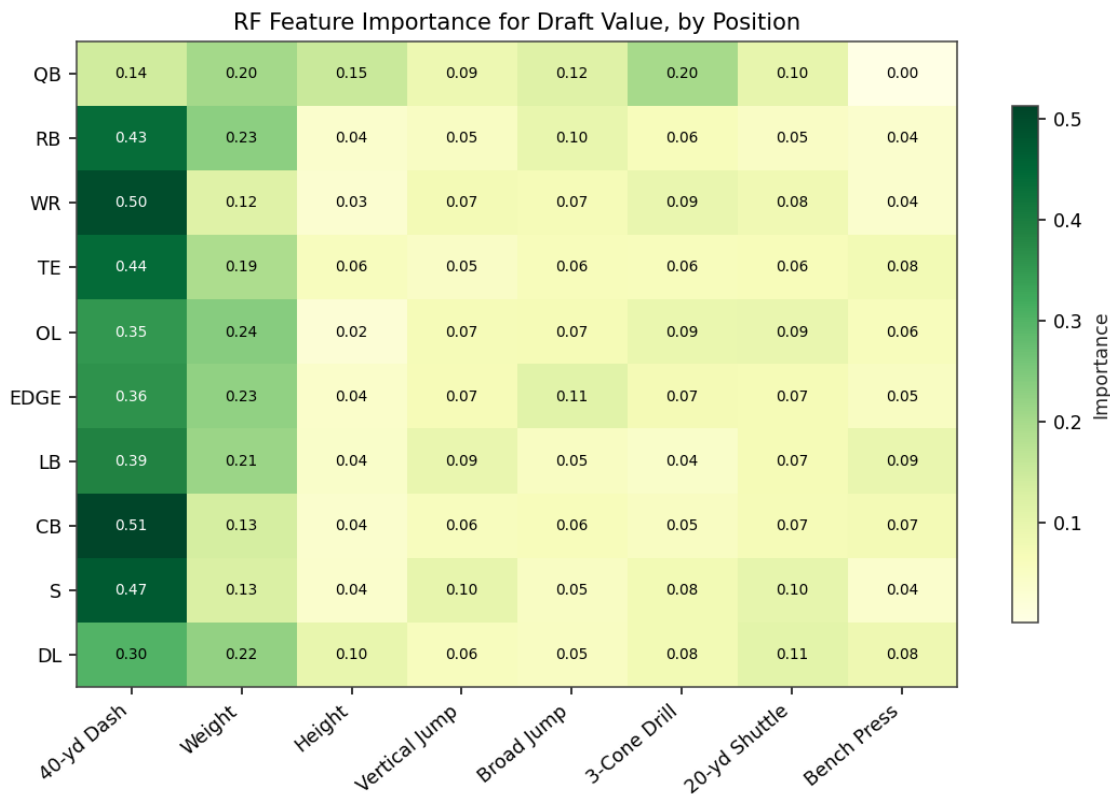
Part 5 — Athletic Outlier Detection

Isolation Forest and Local Outlier Factor are fit separately within each position group (an outlier only means something relative to positional peers), then converted to within-position percentiles for fair cross-position comparison. 'Freak athletes' = unusual AND elite (SAI $\geq$ 90).

Player	Pos	Year	SAI	Unusual %ile
Armand Membou	OL	2025	99.0	87%
Anthony Richardson	QB	2023	98.2	99%
Chris Chambers	WR	2001	97.7	97%
Alvin Dupree	EDGE	2015	96.6	95%
Jarrod Cooper	S	2001	96.4	94%
Marcus Heit	ST	2014	96.3	89%
Jordan Davis	DL	2022	95.9	92%
Jared Retkofsky	ST	2007	95.7	85%
Jake Locker	QB	2011	95.6	86%
Tyler Owens	S	2024	95.6	92%

Part 6 — Which Combine Drills Actually Matter?

A separate Random Forest is fit per position (not one model for everyone), then feature importances are extracted for that position only. This is the most actionable finding in the project:

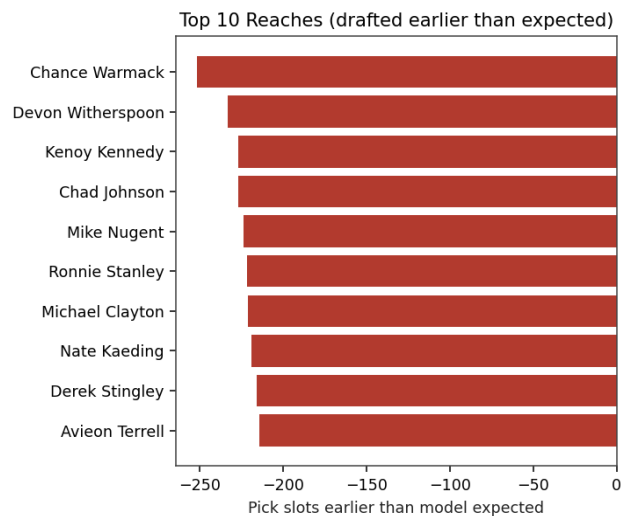
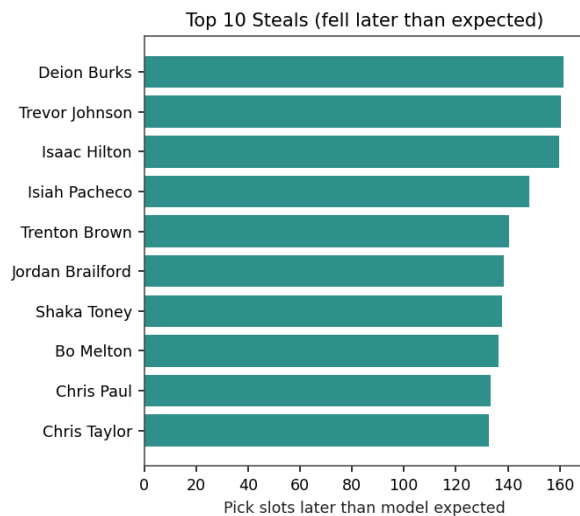


40-yard dash dominates for nearly every position — even OL and DL, where it acts as a proxy for overall explosiveness/athleticism rather than straight-line speed being the skill itself. **Bench press is consistently the weakest predictor** across the board — matching the common scouting critique that bench is more a measure of gym strength than football-relevant power. **Weight matters heavily for OL/DL/TE.**

Part 7 — Draft Steals & Reaches

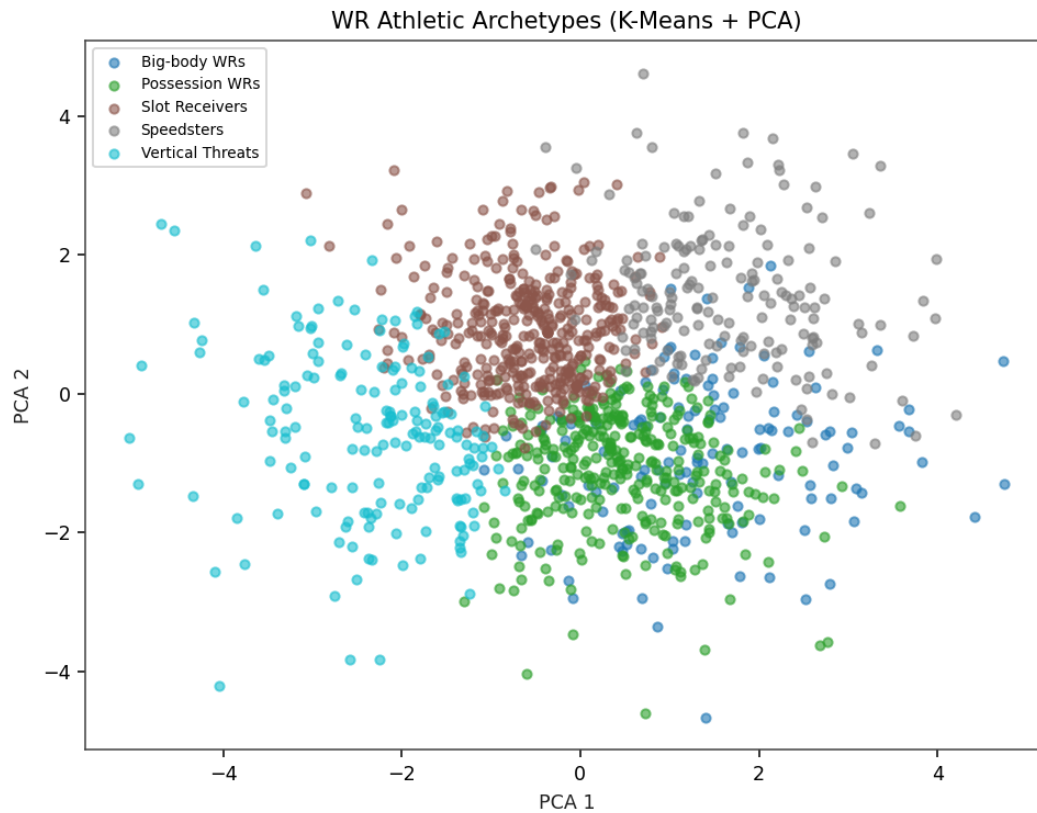
Steal = fell later than a measurables-only model expected. **Reach** = drafted earlier than measurables alone would predict.

Since the model only explains ~20% of draft-position variance, predictions regress hard toward the mean — so top-10 picks will systematically read as 'reaches' and Day 3 picks as 'steals.' This says more about what workout numbers can't capture (tape, production, scheme fit) than about actual scouting quality. Tom Brady is the canonical illustration: his measurables were simply below-average across the board, not predictive of either an all-time great or specifically a Round 6 pick — the model would not have 'found' him from his workout alone.



Part 8 — Position Clustering

K-Means (primary) plus Agglomerative clustering as a structural cross-check, with PCA for 2D visualization (UMAP was unavailable in this offline sandbox; PCA is used instead and noted in the README). Each position gets a curated number of clusters matching how many distinct athletic roles scouts actually use for that position.



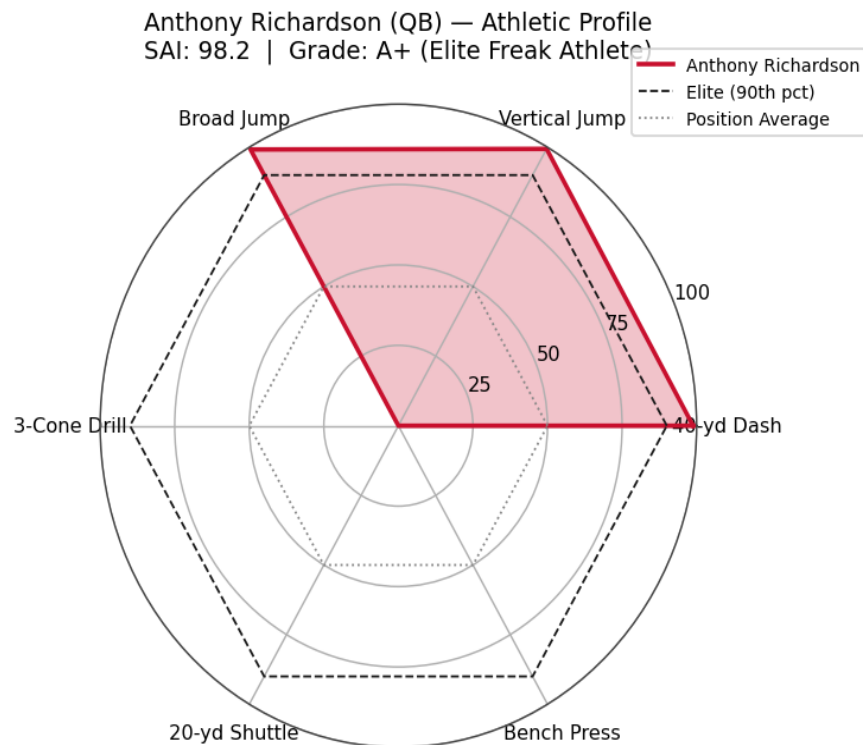
WR archetypes: Speedsters, Possession, Big-body, Slot Receivers, Vertical Threats — named automatically by ranking cluster centroids on speed and size.

Part 9 — Interactive Dashboard

A self-contained static HTML dashboard (`dashboard/index.html`) covers Player Search (auto-generated scouting reports), Position Explorer (benchmarks, drill importance, archetypes), a client-side Draft Predictor, Steals & Reaches, and Outliers. It runs entirely in the browser (no server) against a precomputed data bundle, so nearest-neighbor comparables and percentiles are computed live in JavaScript.

Part 10 — Scouting Report Generator

Given a player name, `src/scouting_report.py` auto-produces an athletic profile, strengths/weaknesses (percentile vs. position peers), closest comparables, model-expected draft range, an SAI letter grade, and a radar chart. Sample below.



Anthony Richardson (QB, 2023) — SAI 98.2, Grade A+ (Elite Freak Athlete). Drafted 4th overall; model-expected range was Round 2 from measurables alone, since his elite testing wasn't matched by elite accuracy on tape — a case where the model and actual value diverge in an informative way.